

7.5 Carboxylic Acids & Derivatives (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	7. Organic Chemistry (A Level Only)
Topic	7.5 Carboxylic Acids & Derivatives (A Level Only)
Difficulty	Medium

Time allowed: 60

Score: /49

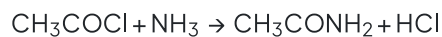
Percentage: /100

Question 1a

Acyl chlorides are commonly used carboxylic acid derivatives.

Ethanoyl chloride reacts with an excess of ammonia.

A student proposes the following equation to represent the reaction:



Correct and explain the student's error.

[2 marks]

Question 1b

N-methylethanamide can also be formed from ethanoyl chloride.

i)

Draw the skeletal formula of *N*-methylethanamide.

[1]

ii)

Write an equation for the formation of *N*-methylethanamide from ethanoyl chloride.

[2]

[3 marks]

Question 1c

Ethanoyl chloride will react with 2-methylpropan-1-ol to form an ester.

Draw the structure of the ester formed.

[1 mark]

Question 2a

Esters can be formed from carboxylic acids and acyl chlorides.

Write an equation for the formation of the ester formed from methanol and propanoic acid.

[1 mark]

Question 2b

Propanoyl chloride can be prepared from propanoic acid.

i)

Write an equation for the preparation of propanoyl chloride from propanoic acid and phosphorus(V) chloride.

[1]

ii)

Give **one** observation of this reaction.

[1]

[2 marks]

Question 2c

Name and outline a mechanism for the reaction between methanol and propanoyl chloride to form methyl propanoate.

[5 marks]

Question 2d

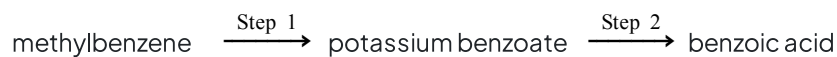
Describe **two** advantages of using propanoyl chloride over propanoic acid in the preparation of esters in industry.

[2 marks]

Question 3a

Benzoic acid can be formed from methylbenzene.

This process occurs in two stages.



i)

Draw the displayed formula for methylbenzene.

[1]

ii)

Draw the displayed formula for benzoic acid.

[1]

[3 marks]

Question 3b

i)

Name the reaction that takes place in **step 1**.

[1]

ii)

Describe the conditions and reagents needed for **step 1** and **2** of this reaction.

Step 1:

Step 2:

[2]

[3 marks]**Question 3c**Describe the observations that could be made during **step 1**.**[2 marks]****Question 3d**

Describe and explain the relative acidities of benzoic acid and phenol.

[3 marks]

Question 4a

This question is about butanoyl chloride.

Butanoyl chloride can be produced from butanoic acid.

- i)
Draw the displayed formula of butanoyl chloride.
- ii)
Give the name and formula of a reagent that produces butanoic acid from butanoyl chloride.

[1]

[2]

[3 marks]**Question 4b**

- i)
Explain why butanoic acid is a weaker acid than 2-chlorobutanoic acid.

[2]

- ii)
Draw the skeletal formula for 2-chlorobutanoic acid.

[1]

[3 marks]**Question 4c**

Name a chloro-substituted butanoic acid that is a stronger acid than 2-chlorobutanoic acid.

[1 mark]

Question 4d

Butanoyl chloride can undergo hydrolysis.

Outline a mechanism for the hydrolysis of butanoyl chloride.

[4 marks]

Question 4e

Explain the relative ease of hydrolysis for butanoyl chloride, chlorobenzene and 1-chlorobutane.

[5 marks]

Question 5a

Methanoic acid, HCO_2H , has a similar acid strength to propanoic acid.

Describe a chemical test to distinguish between these two acids. Name the acid which gives a positive result in this test and describe the observations that would be made.

[2 marks]

Question 5b

The ester phenyl propanoate, $\text{C}_2\text{H}_5\text{CO}_2\text{C}_6\text{H}_5$, can be made from phenol and propanoic acid in a two step synthesis.

The first step produces an acyl chloride.

For this two step synthesis,

- draw the structure of the product of the first step
- state the reagents and conditions needed for each step of the synthesis

[3 marks]

Question 5c

Explain why an acyl chloride should be formed first for the reaction in part (b).

[1 mark]

Model Answers: Medium

1a

a) The correct equation is:

- $\text{CH}_3\text{COCl} + 2\text{NH}_3 \rightarrow \text{CH}_3\text{CONH}_2 + \text{NH}_4\text{Cl}$; [1 mark]
- The excess of ammonia means that a second ammonia molecule reacts with the HCl because it is basic

AND

Ammonium chloride is formed as the second product (not hydrogen chloride); [1 mark]

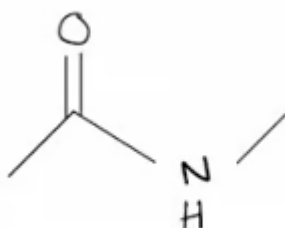
[Total: 2 marks]

- The student's equation would have been correct if **excess** ammonia had not been used

1b

b)

i) The skeletal formula of *N*-methylethanamide is:



- Correct skeletal formula; [1 mark]

ii) An equation for the formation of *N*-methylethanamide from ethanoyl chloride is:



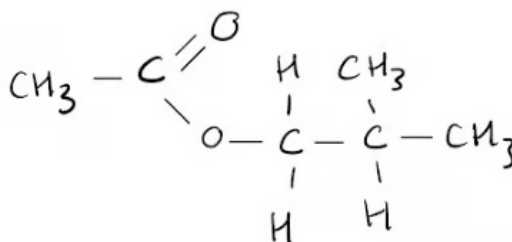
- Correct reactants; [1 mark]
- Correct products and balancing; [1 mark]

[Total: 3 marks]

- **Remember:** The name having the *N* at the beginning indicates that the methyl group is attached to the nitrogen rather than a carbon atom in the main carbon chain
- Once you have drawn your amide you can tell it is a secondary amide as the nitrogen atom is attached to two carbon atoms
- Secondary amides are formed from the reaction between ethanoyl chloride and a primary amine

1c

c) The structure of the ester is:



- Correct displayed formula; [1 mark]

[Total: 1 mark]

- You should be able to draw the structures of esters for any combination of alcohol and carboxylic acid / acyl chloride / acid anhydride

2a

a) An equation for the formation of the ester formed from methanol and propanoic acid is:

- $\text{CH}_3\text{OH} + \text{CH}_3\text{CH}_2\text{COOH} \rightarrow \text{CH}_3\text{CH}_2\text{COOCH}_3 + \text{H}_2\text{O}$; [1 mark]

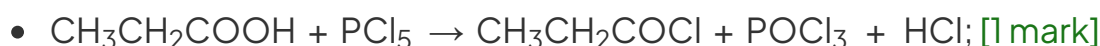
[Total: 1 mark]

- To identify the ester produced:
 - The first part of the ester name comes from the alcohol so **methanol** = **methyl**
 - The second part of the name comes from the carboxylic acid so **propanoic** = **propanoate**
 - Methyl propanoate is formed
- During the reaction, a water molecule is lost and an ester linkage, $-\text{COO}-$ forms between the two groups
 - For this reason it is known as a condensation reaction
- To determine the structural formula, the left hand side of the **$\text{CH}_3\text{CH}_2\text{COO}$** is from the carboxylic acid and the right hand side from the alcohol **$\text{CH}_3\text{CH}_2\text{COOCH}_3$**

2b

b)

i) An equation for the preparation of propanoyl chloride from propanoic acid and phosphorus(V) chloride is:



ii) One observation of this reaction is:

- White fumes (of HCl); [1 mark]

[Total: 2 marks]

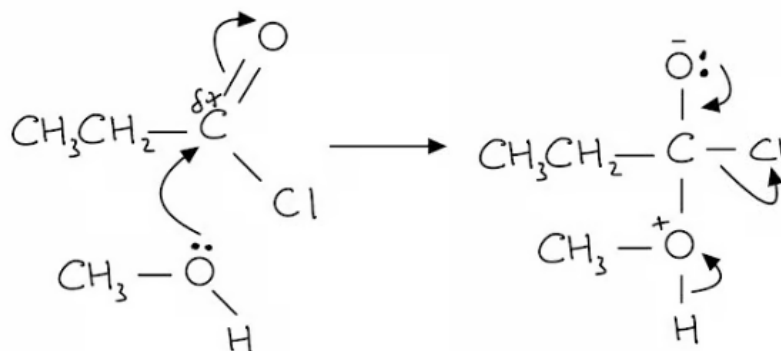
- Acyl chlorides are prepared from their corresponding carboxylic acid
 - So **propanoyl** chloride is produced from **propanoic** acid and so on
- Three different reagents can be used in this preparation, these reagents are PCl_5 , PCl_3 , and SOCl_2
- You should be able to write equations for the preparation of an acyl chloride using all three

2c

c) Name and outline a mechanism for the reaction between methanol and propanoyl

chloride to form methyl propanoate:

- Mechanism: (Nucleophilic) Addition- Elimination; [1 mark]



- Curly arrow from the lone pair on the oxygen (of the OH) to carbon atom (of the C=O)
AND
 $\delta+$ on carbon atom; [1 mark]
- Curly arrow from C=O bond to the O of the C=O bond; [1 mark]
- Three arrows correct
AND
Lone pair on the O atom; [1 mark]
- Correct structure of ion including the two charges; [1 mark]

[Total: 5 marks]

- The alcohol acts as a nucleophile in this reaction
- The lone pair on the oxygen attacks the carbonyl carbon, and an HCl molecule is eliminated
- Make sure your curly arrows are clear, and that they are double headed

2d

d) Two advantages of using propanoyl chloride over propanoic acid in the preparation of esters in industry are:

- Curly arrow from the lone pair on the oxygen (of the OH) to carbon atom (of the C=O)

AND

$\delta+$ on carbon atom; [1 mark]

- Curly arrow from C=O bond to the O of the C=O bond; [1 mark]
- Three arrows correct

AND

Lone pair on the O atom; [1 mark]

- Correct structure of ion including the two charges; [1 mark]

[Total: 5 marks]

- The alcohol acts as a nucleophile in this reaction
- The lone pair on the oxygen attacks the carbonyl carbon, and an HCl molecule is eliminated
- Make sure your curly arrows are clear, and that they are double headed

2d

d) Two advantages of using propanoyl chloride over propanoic acid in the preparation of esters in industry are:

- The reactions with acyl chlorides are quicker; [1 mark]
- The acyl chloride reactions also go to completion and do not form an equilibrium mixture; [1 mark]

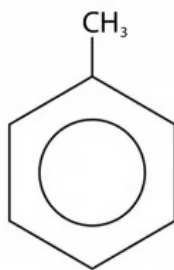
[Total: 2 marks]

- A reaction that forms an equilibrium mixture will end up with a lower yield of product due to the reverse reaction continuing to make reactants

3a

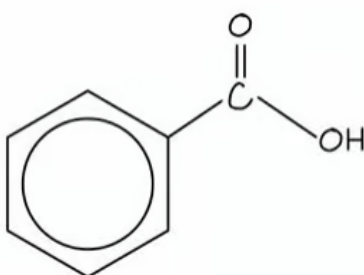
a)

i) The displayed formula of methylbenzene is:



- Correct structure: [1 mark]

ii) The displayed formula of benzoic acid:



- Correct structure; [1 mark]

[Total: 2 marks]

3b

b)

i) The reaction that takes place in **step 1** is:

- Oxidation; [1 mark]

ii) The conditions and reagents needed for **step 1** and **2** of this reaction are:

- Step 1: Heated / reflux
AND
Hot alkaline KMnO₄; [1 mark]
- Step 2: dilute acid; [1 mark]

[Total: 3 marks]

- You would score mark 3 for naming a specific acid such as dilute HCl
- In step 2, the potassium benzoate is protonated using the dilute acid to produce benzoic acid

3c

c) The observations that could be made are:

- Solution changes from purple to colourless; [1 mark]
- A brown precipitate forms; [1 mark]

[Total: 2 marks]

- KMnO_4 , the oxidising agent used in step 1, is a purple colour due to the presence of Mn^{7+} ions
- The oxidising agent is itself reduced, with the Mn^{7+} ions being reduced to Mn^{4+}
- As this happens, the purple colour disappears to give a colourless solution
- A brown precipitate of MnO_2 also forms

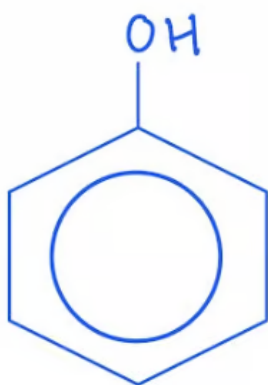
3d

d) Describe and explain the relative acidities of benzoic acid and phenol:

- Benzoic acid is a stronger acid than phenol; [1 mark]
- (Because) the carbonyl group withdraws electron density away from the O-H bond; [1 mark]
- So it is more likely to release / lose a H^+ / proton; [1 mark]

[Total: 3 marks]

- Phenol has the structure:

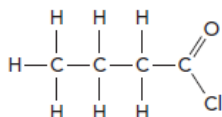


- **Remember:** The stronger the acid, the more easily it loses a proton
- The presence of the carbonyl group in benzoic acid enables electron density to be drawn away from the O-H bond, weakening it, and causing easier release of the H^+
- Phenol does not contain a carbonyl group so it is hard to lose the H^+

4a

a)

i) The displayed formula of butanoyl chloride is:



- Correct displayed formula; [1 mark]

ii) The name and formula of a reagent that produces butanoic acid from butanoyl chloride is:

Alternative 1:

- PCl_5
AND
phosphorus(V) chloride; [1 mark]

- Correct displayed formula; [1 mark]

ii) The name and formula of a reagent that produces butanoic acid from butanoyl chloride is:

Alternative 1:

- PCl_5
AND
phosphorus(V) chloride; [1 mark]

Alternative 2:

- PCl_3
AND
phosphorus(III) chloride; [1 mark]

Alternative 3:

- SOCl_2
AND
sulfur dichloride oxide; [1 mark]

[Total: 2 marks]

- Acyl chlorides have the functional group $-\text{COCl}$
- Butanoyl chloride has four carbon atoms, the fourth being that in the $-\text{COCl}$ group
- You might have to write equations for the reaction between a carboxylic acid and a specific reagent so you must learn the names and formulas for all three

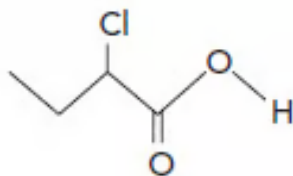
4b

b)

i) Butanoic acid is a weaker acid than 2-chlorobutanoic acid because:

- Chlorine atom withdraws electrons from the $-\text{COOH}$ group; [1 mark]
- Weakening the $-\text{OH}$ bond so it is easier for 2-chlorobutanoic acid to lose an H^+ ion; [1 mark]

ii) The skeletal formula for 2-chlorobutanoic acid is:



- Correct skeletal formula; [1 mark]

[Total: 3 marks]

- Acids lose a proton, H^+ in aqueous solution
- The more easily the H^+ ion is lost the stronger the acid
- Butanoic acid contains a carbonyl group which draws electron density away from the O-H bond in the $-COOH$ group making it relatively easy to lose a proton
- 2-chlorobutanoic acid contains a carbonyl group **AND** a chlorine atom (attached to the carbon atom which is bonded to the carbonyl group)
 - Chlorine is also able to withdraw electron density away from the O-H in the $-COOH$ group making the bond even weaker than that of butanoic acid and making it easier to lose the H^+

4c

c) A chloro-substituted butanoic acid that is a stronger acid than 2-chlorobutanoic acid is:

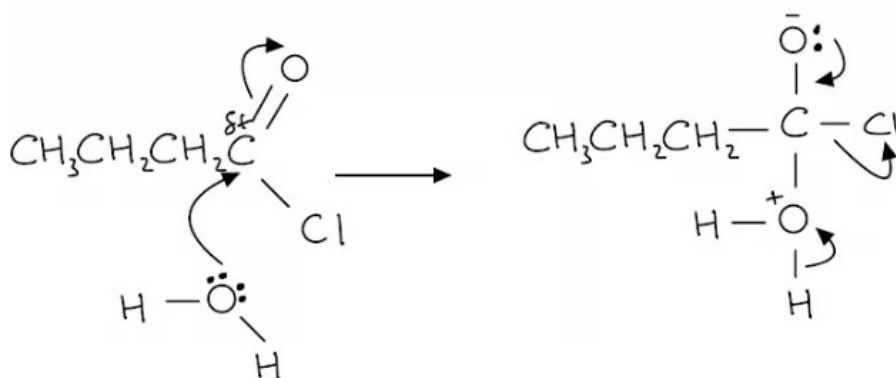
- 2,2-dichlorobutanoic acid
OR
2,3-dichlorobutanoic acid; [1 mark]

[Total: 1 mark]

- The more chlorine atoms there are in the carboxylic acid, the stronger the acid is
- This is because they can draw more electron density away from the O-H bond, weakening it and causing it to more readily lose H^+

4d

d) The mechanism for this reaction is:



- Curly arrow from the lone pair on the oxygen (of the H_2O) to carbon atom (of the $\text{C}=\text{O}$)
AND
 δ^+ on carbon atom; [1 mark]
- Curly arrow from $\text{C}=\text{O}$ bond to the O of the $\text{C}=\text{O}$ bond; [1 mark]
- Three arrows correct
AND
Lone pair on the O atom; [1 mark]
- Correct structure of ion including the two charges; [1 mark]

[Total: 5 marks]

- It is definitely worth learning this addition-elimination mechanism, as it crops up a lot
- Make sure your curly arrows are double headed and you have the δ^+ on the carbon atom

4e

e) Explain the relative ease of hydrolysis for butanoyl chloride, chlorobenzene and 1-chlorobutane:

- The ease of hydrolysis with that most easily broken down is butanoyl chloride > 1-chlorobutane > chlorobenzene; [1 mark]

Explanation for order of butanoyl chloride and 1-chlorobutane:

- (Butanoyl chloride undergoes hydrolysis more easily than 1-chlorobutane because) butanoyl chloride has a carbon atom bonded to two electronegative atoms / oxygen as well as chlorine; [1 mark]

- So the carbon atom is more strongly δ^+ than the carbon atom in 1-chlorobutane; [1 mark]

Explanation for chlorobenzene:

- A lone pair of electrons from the Cl atom overlaps with the delocalised p electrons in the benzene ring; [1 mark]
- Making the C-Cl bond stronger / harder to break; [1 mark]

[Total: 5 marks]

- Acyl chlorides can be hydrolysed at room temperature using a neutral water molecule as a nucleophile
- Alkyl chlorides such as 1-chlorobutane, require a strong alkali to be refluxed with it to provide a source of :OH^- ions
 - The OH^- ions are a stronger nucleophile than H_2O
- Aryl chlorides, like chlorobenzene, have the carbon atom bonded to the chlorine atom as part of the delocalised π bonding system of the benzene ring
 - One of the lone pairs of electrons of the Cl atom overlaps with this delocalised system
 - The C-Cl bond, therefore, has some double-bond character causing it to become stronger and harder to break

5a

a) A chemical test to distinguish between the two acids could be:

- Tollens' reagent
OR
Fehling's reagent / solution; [1 mark]
- Methanoic acid gives a silver mirror/solid with Tollens' reagent
OR
Red / orange ppt / solid with Fehling's reagent / solution; [1 mark]

[Total: 2 marks]

- Tollens' reagent and Fehling's solution will react with methanoic acid as they are

a) A chemical test to distinguish between the two acids could be:

- Tollens' reagent

OR

Fehling's reagent/ solution; [1 mark]

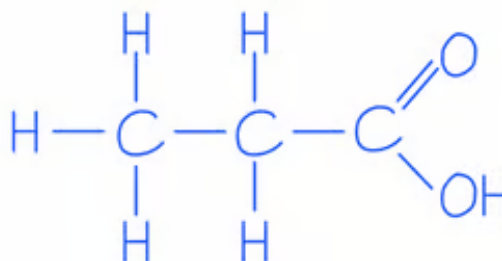
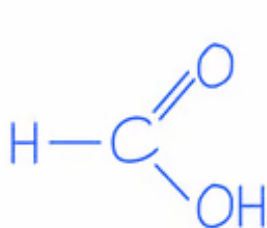
- Methanoic acid gives a silver mirror/solid with Tollens' reagent

OR

Red / orange ppt / solid with Fehling's reagent / solution; [1 mark]

[Total: 2 marks]

- Tollen's reagent and Fehling's solution will react with methanoic acid as there is a carbon bonded to a hydrogen atom and carbon also forms a double bond with oxygen
 - This will react in the same way an aldehyde group will and give a positive result with a Tollens' reagent and Fehling's solution



- This reagent does not react with propanoic acid because even though carbon is doubly bonded to oxygen, it does not have a hydrogen attached to it

5b

b) The steps are:

- PCl_5

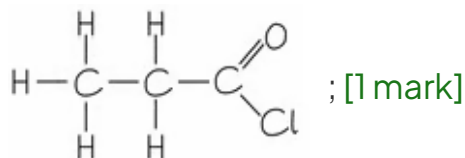
OR

PCl_3 (+ heat)

OR

SOCl_2 (added to propanoic acid); [1 mark]

- Product of first step is:



- Add product of first step to phenol in NaOH; [1 mark]

[Total: 3 marks]

- In order to form an acyl chloride we can react propanoic acid with either
 - Solid phosphorus(V) chloride (PCl_5)
 - Liquid phosphorus(III) chloride (PCl_3) and heat
 - Liquid sulfur dichloride oxide (SOCl_2)
- Acyl chlorides are more reactive than their corresponding carboxylic acids and are therefore often used as starting materials in the production of organic compounds such as esters
- Propanoyl chloride can react phenol to form esters
 - With phenols, the reaction requires heat to proceed and needs to be carried out in the presence of a base
 - The base deprotonates the phenol to form a phenoxide ion which is a better nucleophile than the phenol molecule

5c

c) Any acyl chloride is required because:

- Faster reaction; [1 mark]
- Reaction of phenol and carboxylic acid / propanoic acid does not go to completion; [1 mark]

[Total: 2 marks]

- Carboxylic acids are less reactive than acyl chlorides so the reaction will be slower
- Acyl chlorides are therefore more useful than carboxylic acids in the synthesis of esters as this reaction goes to completion
 - The reaction with carboxylic acids is reversible